

2D and 3D representations and molecular descriptors

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CAMLC | June 2nd 2026



Featurization in (Data)-Chemistry

Outline:

Introducing the terminology

Digitizing chemistry: How do we convert molecules into numbers?

Molecular Descriptors in Different Dimensions

Practical Application using Python

By the end of this lecture you should be able to appreciate some different strategies to featurize molecules, appraise what types of features are applicable to a given problem, and use molecular descriptors to perform a regression task for a target molecular property.

House Price Predictions

Target variable

House price

221,900

Feature vector

Bedrooms	Sqft_living	Floors	Bathrooms
3	1180	1	1

Featurization involves the conversion of raw data into a numerical vector (*feature vector*)

We can then attempt to find a mathematical relationship between the target variable and the feature vector

In the case of Multivariate Linear Regression (MLR)

$$y = m_0x_0 + m_1x_1 + m_2x_2 + m_3x_3 + c$$

House Price Predictions (again!)

Target variable

House price

221,900

Feature vector

Bedrooms	Sqft_living	Floors	Bathrooms	Num. rooms
3	1180	1	1	8

What attributes or properties are used to make up the feature vector?

Feature Engineering often involves the use of domain knowledge – i.e. which attributes are relevant?

What is the best representation of the sample data to learn a solution to the problem?

Feature selection then automatically selects a subset that are most useful to the problem

Some may be dropped due to low statistical importance or correlation with other features.

$$y = m_0x_0 + m_1x_1 + m_3x_3 + m_4x_4 + c$$

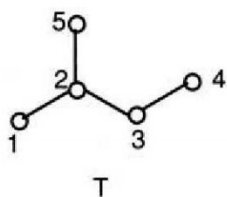
Back to Chemistry: Quantitative Structure-Property Relationships (QSPR)

Boiling Point	molecule	Mol. weight	Weiner	Zagreb1	Zagreb2
36.1	pentane	72.15	20	14	12
68.8	hexane	86.18	35	18	16
98.5	heptane	100.20	56	22	22
125.6	octane	114.23	84	26	24

$$\text{Predicted BP} = 4.4325 * \text{MW} - 0.6411 * \text{Weiner} - 4.3920 * Z1 + 0.2982 * Z2 + 55.5695 \quad (R^2 = 0.99)$$

Molecular Descriptors numerically describe the shapes, electronics, energies of molecules

(a) A labeled tree T



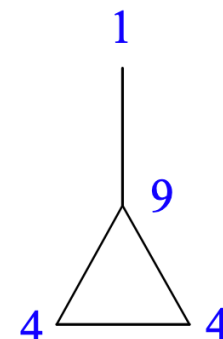
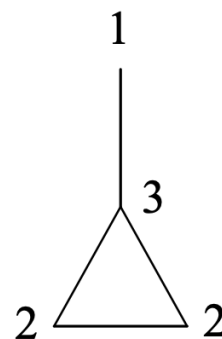
(b) The distance matrix of T

$$\mathbf{D}(T) = \begin{bmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 2 & 1 \\ 2 & 1 & 0 & 1 & 2 \\ 3 & 2 & 1 & 0 & 3 \\ 2 & 1 & 2 & 3 & 0 \end{bmatrix}$$

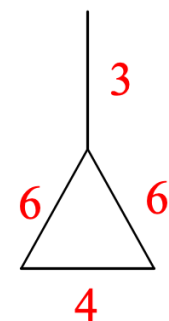
(c) The Wiener number of T

$$W(T) = \frac{1}{2}(8 \cdot 1 + 8 \cdot 2 + 4 \cdot 3) = 18$$

e.g. rdkit, mordred

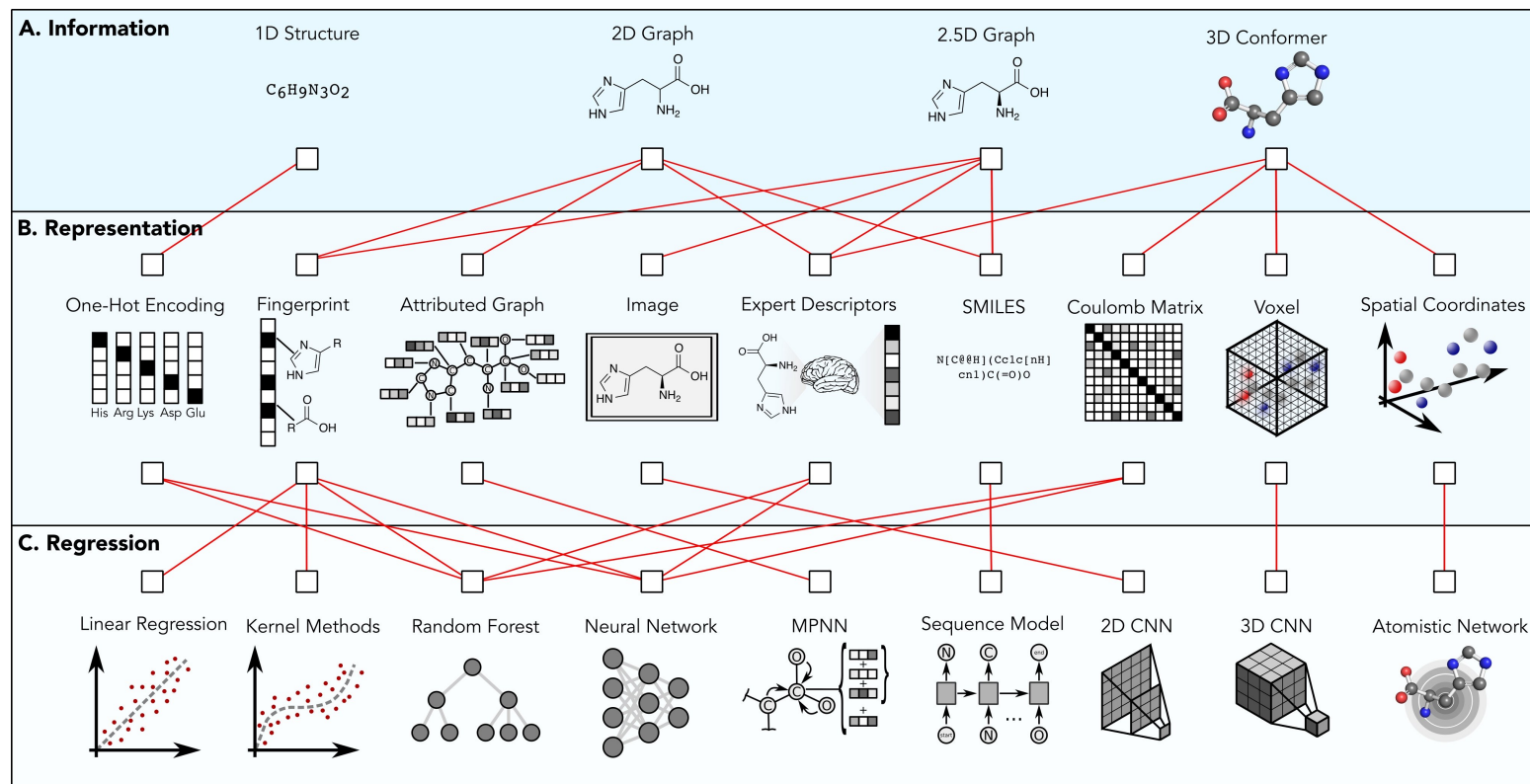


$$M_1 = 18$$



$$M_2 = 19$$

Representing Molecules



Molecular Descriptors

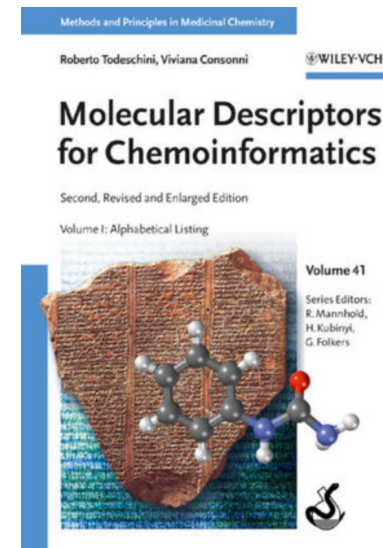
“The molecular descriptor is the final result of a logic and mathematical procedure which transforms chemical information encoded within a symbolic representation of a molecule into a useful number, or the result of some standardized experiment.”

Considerations for Molecular Descriptors

- Interpretability
- Good correlation with at least one property
- No trivial correlation with other molecular descriptors
- Gradual change in its values with gradual changes in the molecular structure
- Insensitive to how collected
- No restrictions to type of molecule
- if possible, discrimination between structural and stereo-isomers.
- Independent of other molecular descriptors

Consonni and Todeschini, Wiley 2009

Head-Gordon, T. Learning to Make Chemical Predictions: The Interplay of Feature Representation, Data, and Machine Learning Methods
Chem, 2020; 6, 1527-1542



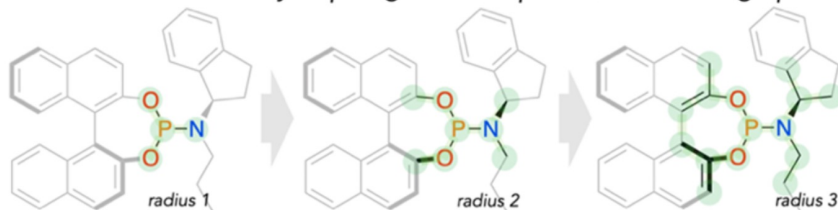
1D representations

Constitutional descriptors

- Num. C atoms = 32 Num. Rings = 7

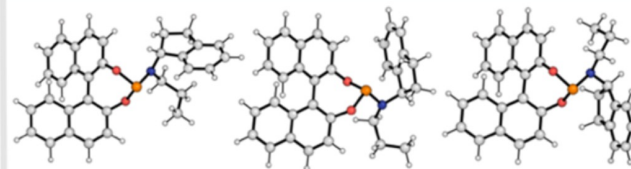
2D representations

Atomic connectivity: topological descriptors & molecular graphs

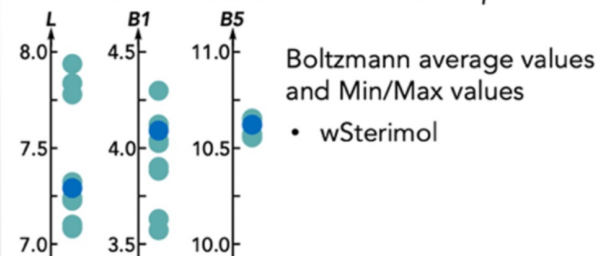


- Morgan/circular fingerprints encode molecular fragments

4D representations



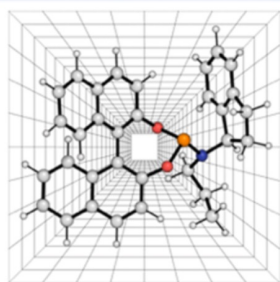
Conformational ensemble descriptors



Boltzmann average values
and Min/Max values

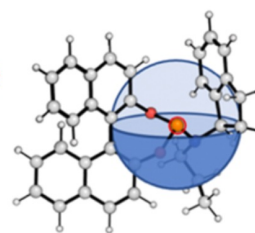
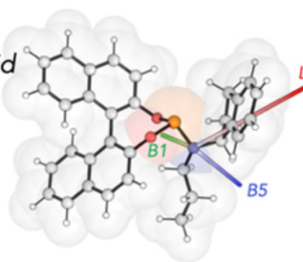
- wSterimol

3D representations



*Grid-based descriptors
map properties to 3D-grid
points (orientation
dependent)*

- CoMFA
- Average Steric
Occupancy (ASO)

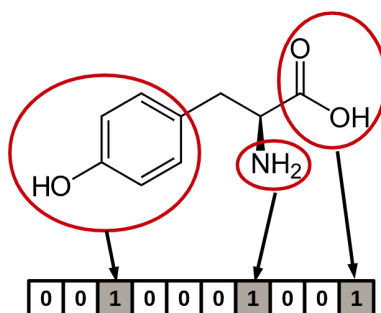


*Optimized 3D-geometry
described by various
shape/steric descriptors:*

- Sterimol (multi-dimensional)
- Buried volume (scalar)

2D Molecular Descriptors: Fingerprints

substructure fingerprint
e.g. MACCS

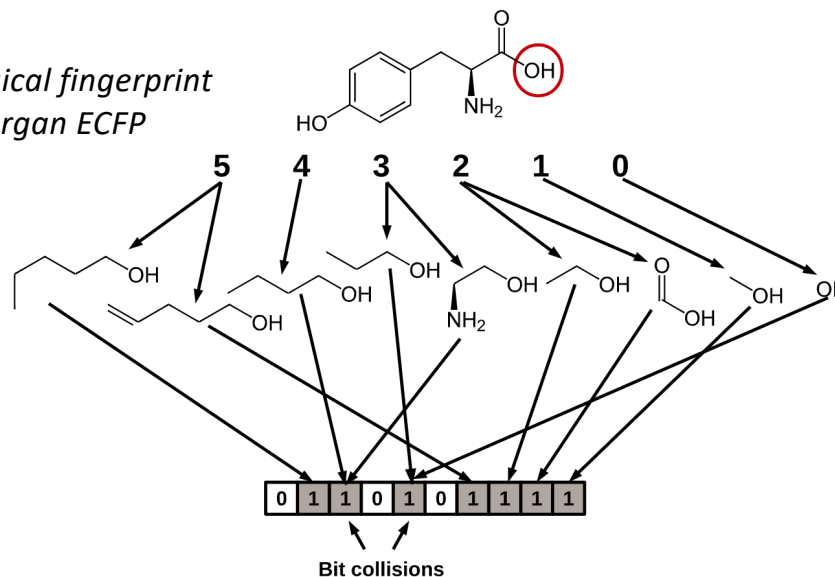


Feature vector

Hypothetical 10-bit substructure fingerprint, with three bits set because the substructures they represent are present in the molecule (circled)

```
fp = mfpgen.GetFingerprint(mol)
print(f'fp: {len(fp)}')
fp: 2048
fp = mfpgen.GetFingerprint(mol, fromAtoms=[1,2,3])
```

topological fingerprint
e.g. Morgan ECFP

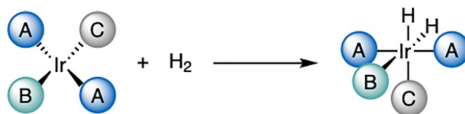


Hypothetical 10-bit topological fingerprint, in this case a linear path-based fingerprint with fragments up to a length of 5.

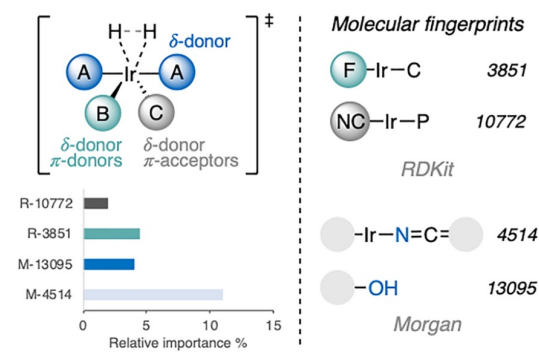
There are two bit collisions, which are likely in fingerprints with a reduced number of bits.

Methods **2015**, 71, 58–63; Count-based is preferable (although more memory required!); *Environ. Sci. Technol.* **2023**, 57, 18193–18202

Aspuru-Guzik and Balcells 2020



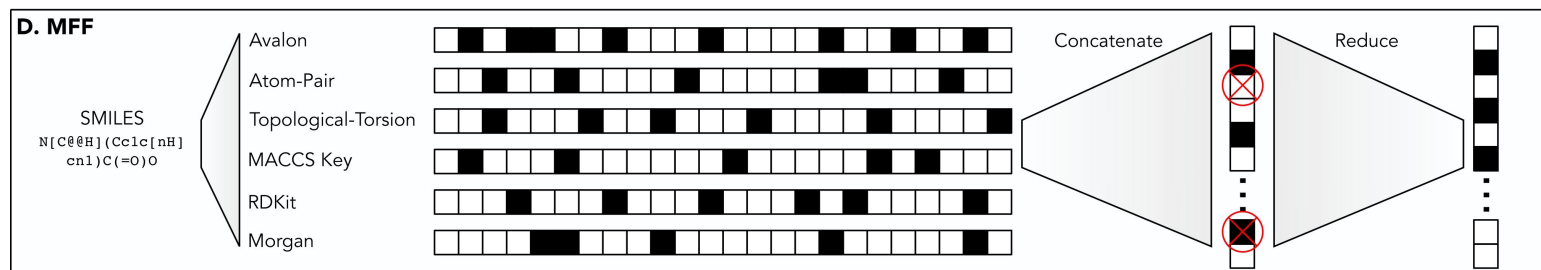
Optimal structure based on topological feature importance



2D Molecular Descriptors: Fingerprints

2500 barrier heights
Regression with a Gaussian process (GP) model
gave an MAE = 0.6 kcal/mol, $R^2 = 0.95$

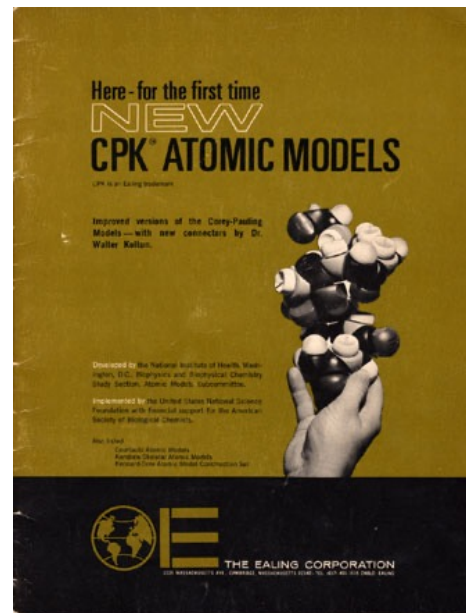
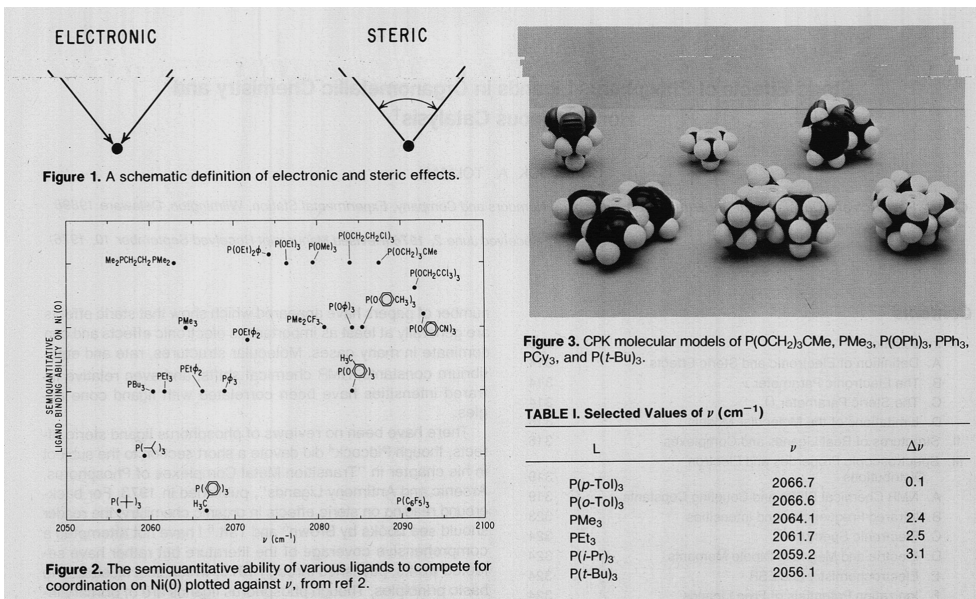
Aspuru-Guzik & Balcells: *Chem. Sci.* **2020**, *11*, 4584–4601



Concatenation of multiple fingerprints used to predict selectivities and yields

Glorius: *Chem.* **2020**, *6*, 1379–1390

3D Molecular Descriptors



Steric Effects of Phosphorus Ligands

Chemical Reviews, 1977, Vol. 77, No. 3 315

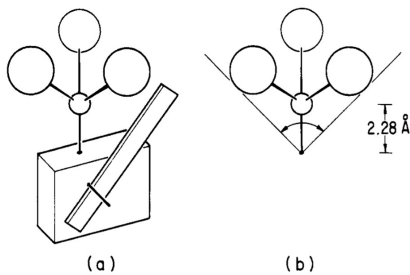


Figure 4. (a) Ligand angle measuring device; (b) the cone angle, from ref 2.

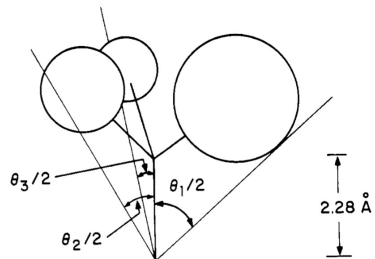
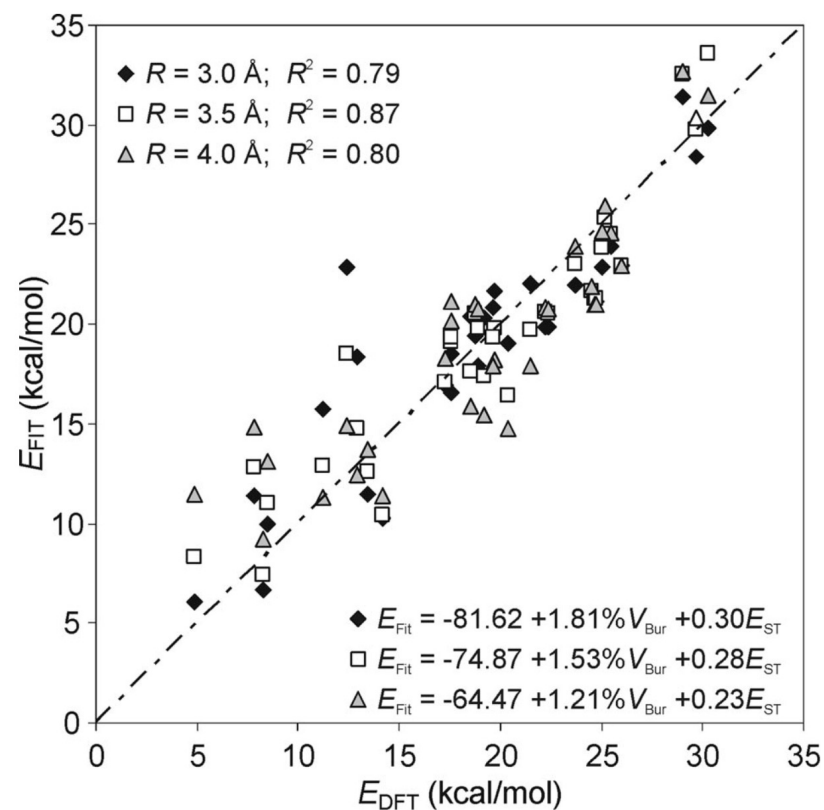
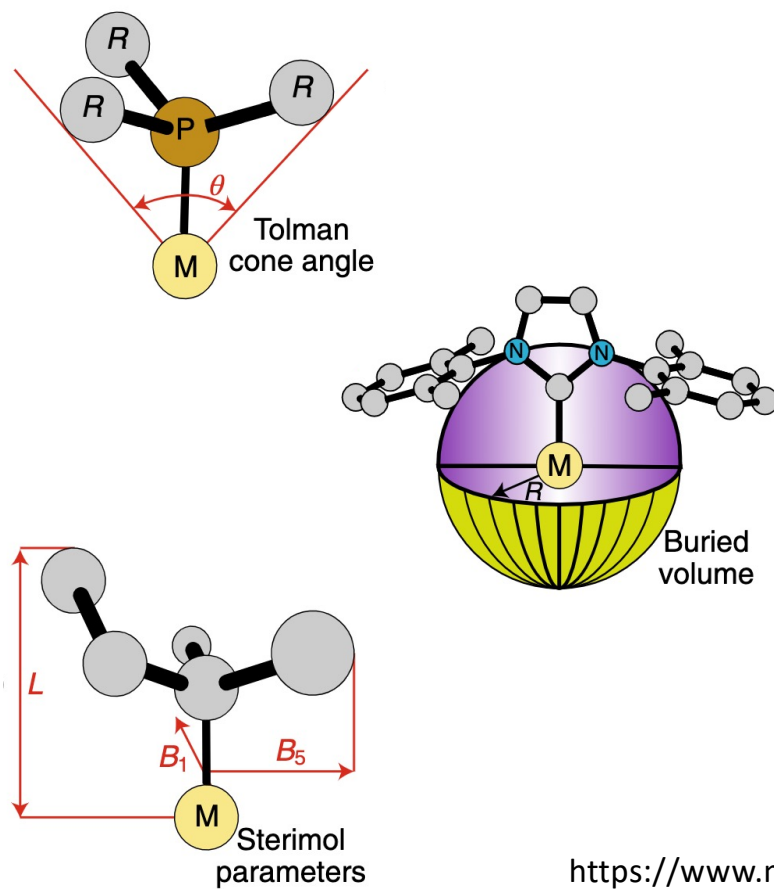


Figure 6. Method of measuring cone angles for unsymmetrical ligands.

Corey-Pauling-Koltun space-filling model

Tolman Chem Rev. 1977, 77, 315

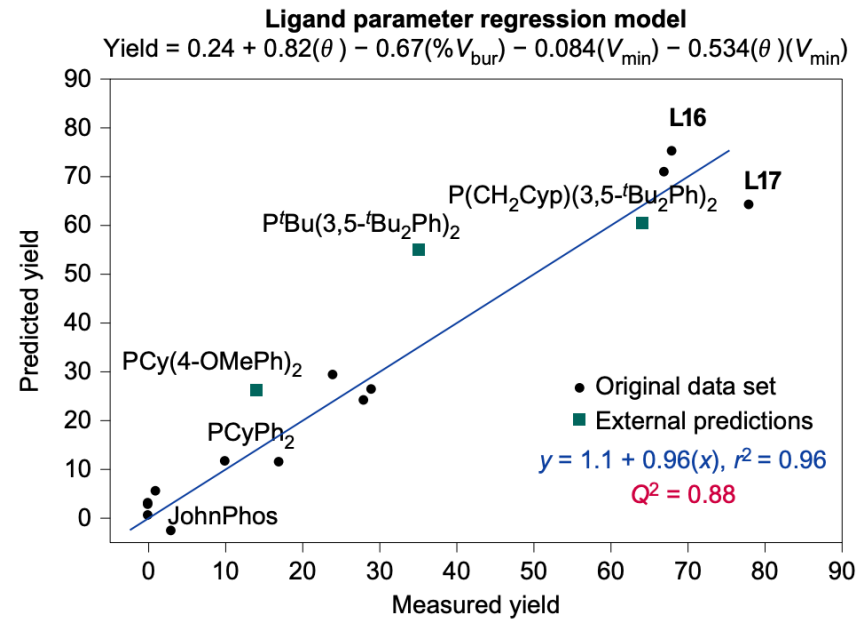
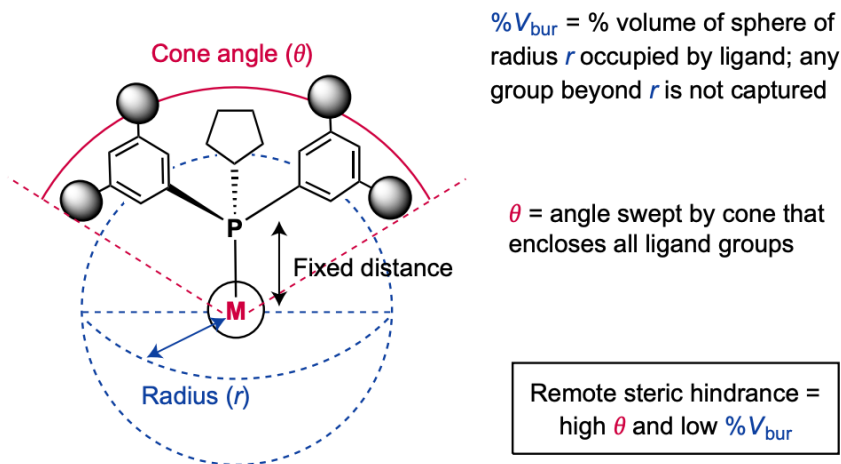
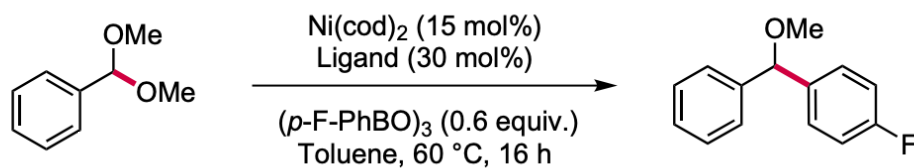
3D Molecular Descriptors: Buried Volume



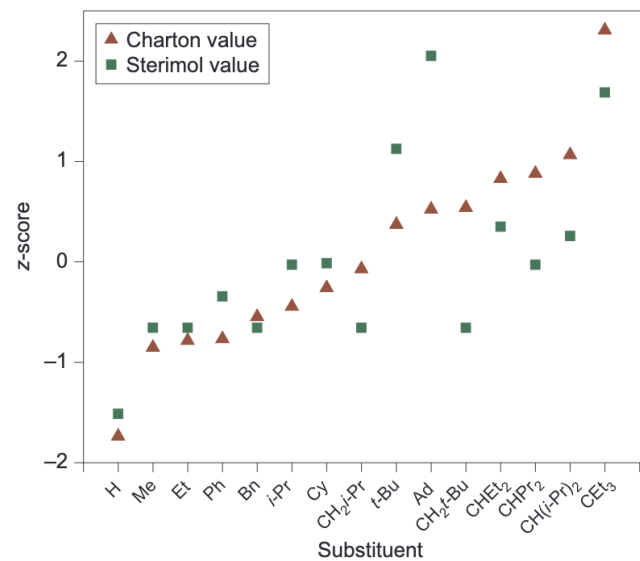
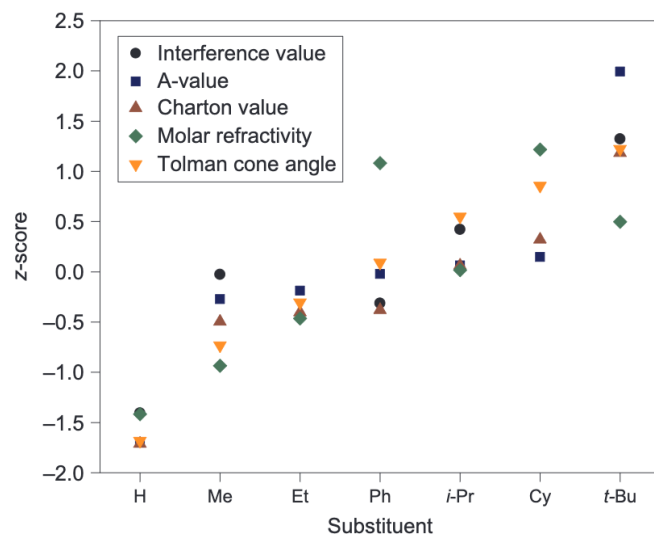
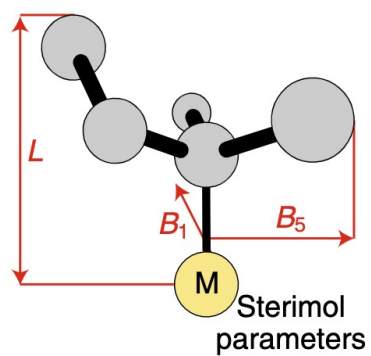
<https://www.molnac.unisa.it/OMtools/sambvca2.1/index.html>

Cavallo & Nolan *Eur. J. Inorg. Chem.* **2009**, 1759–1766; *Nat. Chem.* 2019, 11, 872 12

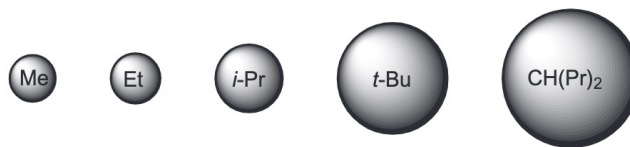
3D Molecular Descriptors: Buried Volume



3D Molecular Descriptors: Sterimol

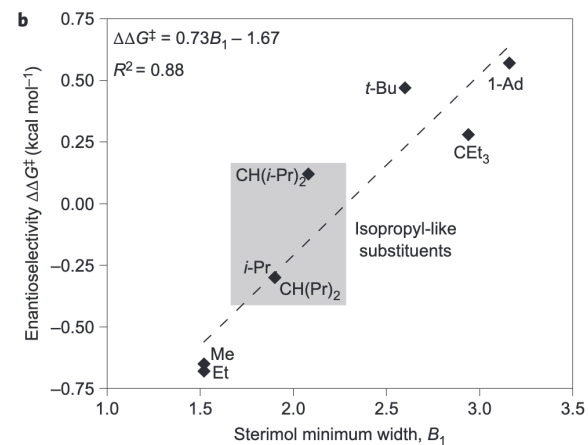
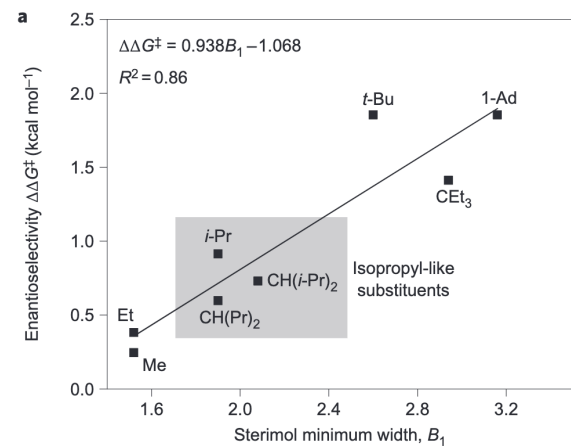
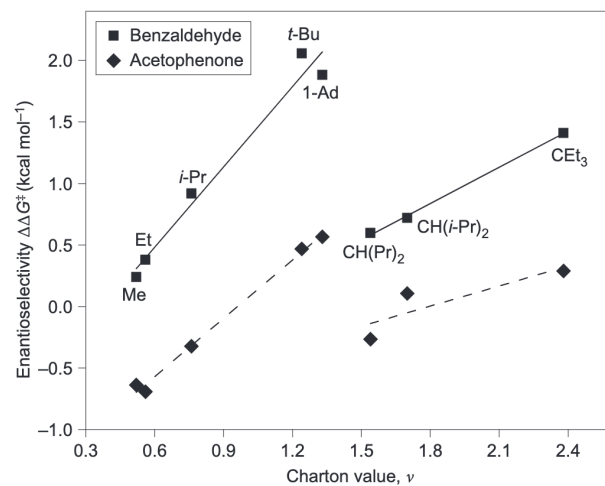
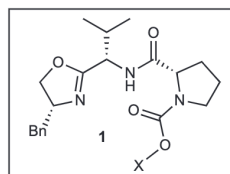
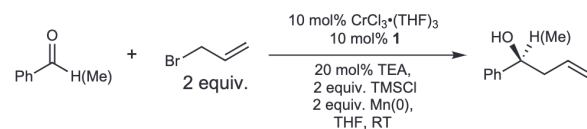
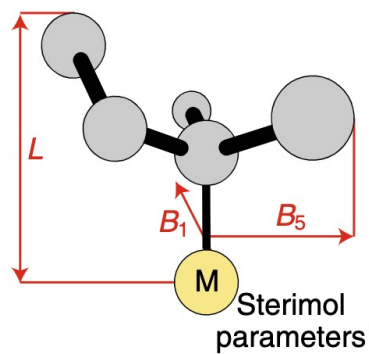


◆ Scaled sphere approximations



Sigman Nat. Chem. **2012**, 4, 366

3D Molecular Descriptors: Sterimol



3D Molecular Descriptors

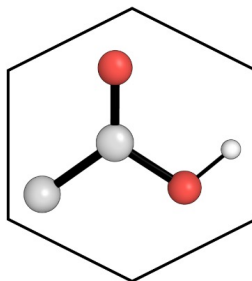


Descriptor Libraries

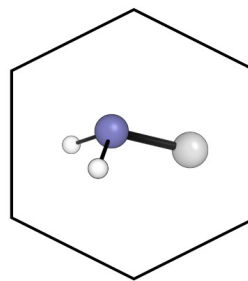
The Descriptor Libraries project is a collaborative effort between The Molecular Sciences Software Institute (MolSSI) and the Center for Computer Assisted Synthesis (C-CAS), both funded by the National Science Foundation. The Descriptor Libraries project combines molecular libraries for various chemical applications from C-CAS in an accessible web application and software framework developed by the MolSSI.



kraken



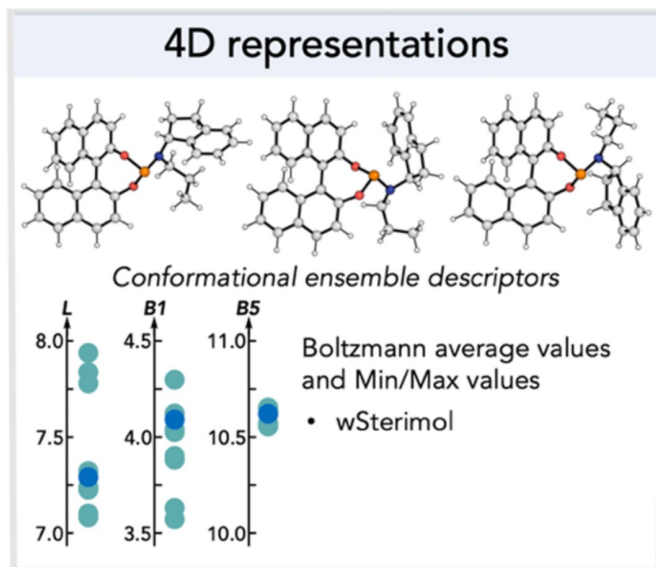
Carboxylic Acids



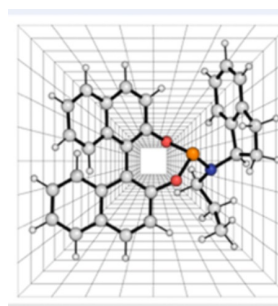
Primary Alkyl Amines

<https://descriptor-libraries.molssi.org/> | <https://autoqchem.org/>

4D Molecular Descriptors?



- Boltzmann average
- Most stable conformer
- Minimum/Maximum values

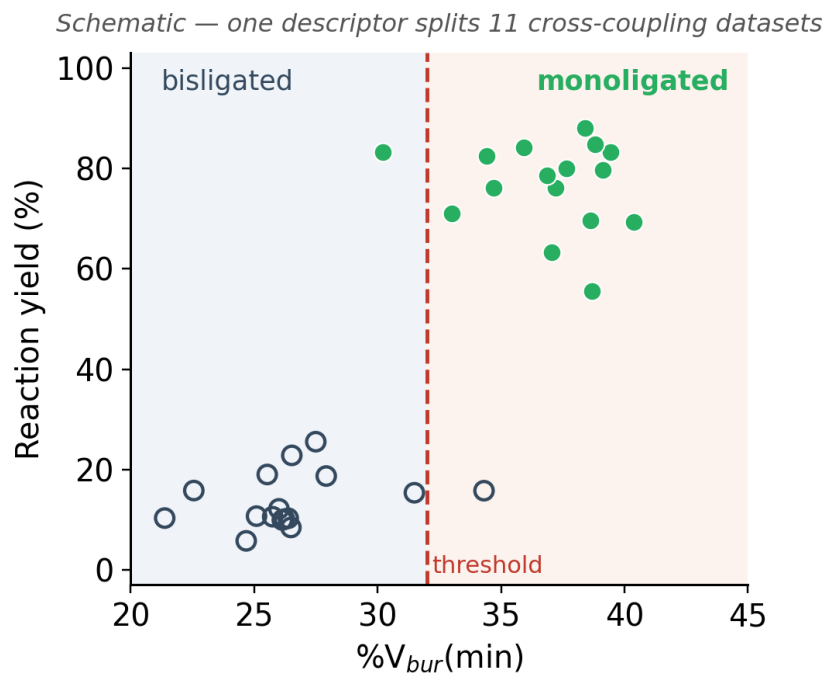


- COMFA / Grid QSAR
- Alignment problem

3D Molecular Descriptors: Reactivity Cliffs

One interpretable 3D descriptor with a decision boundary

- %V_{bur}(min) — the smallest %buried volume across the whole conformer ensemble, not just the lowest-energy conformer.
- A single threshold value splits 11 Ni-/Pd-catalysed cross-coupling datasets into active vs inactive regions.
- The threshold is physically meaningful: it marks the switch between bis- and mono-ligated metal.
- Conformer-ensemble descriptors come straight from the kraken library ($\approx 1,558$ organophosphorus ligands).



Three Eras of Molecular Representation

Same regression target — three answers to “where do the features come from?”

1 • Engineered descriptors

Pick features by hand from domain knowledge.

*MW · logP · Sterimol · %Vbur · MACCS /
Morgan fingerprints*

+ interpretable; strong in low-data regimes

– needs insight; may miss the key axis

2 • Supervised end-to-end

Learn the embedding from the graph for one task.

*Message-passing GNNs — ALFABET,
Chemprop*

+ no manual featurization; SOTA with data

– data-hungry; one model per property

3 • Self-supervised pretraining

Pretrain on millions of unlabelled molecules, then fine-tune.

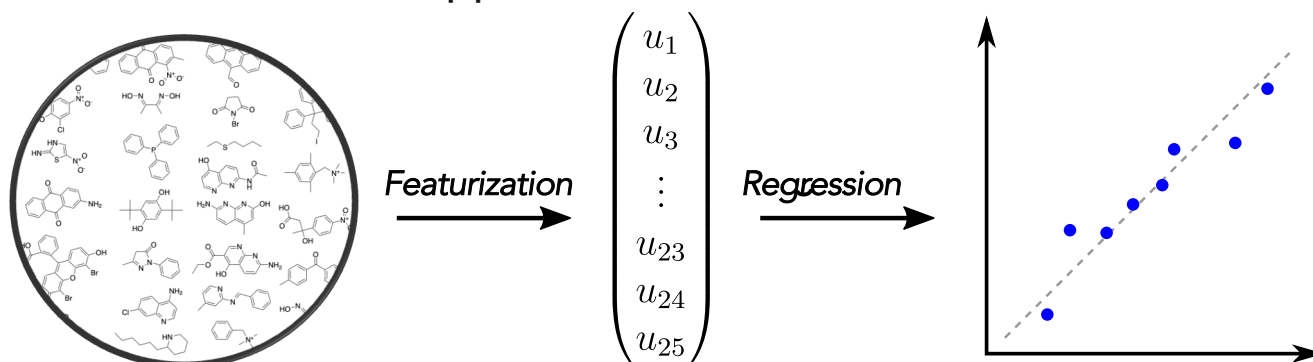
*Foundation models — Uni-Mol (3D),
MolFormer, MoleCLIP*

+ data-efficient transfer; robust to shift

– large, opaque; pretraining cost

hand-crafted → learned per task → learned once, transferred everywhere

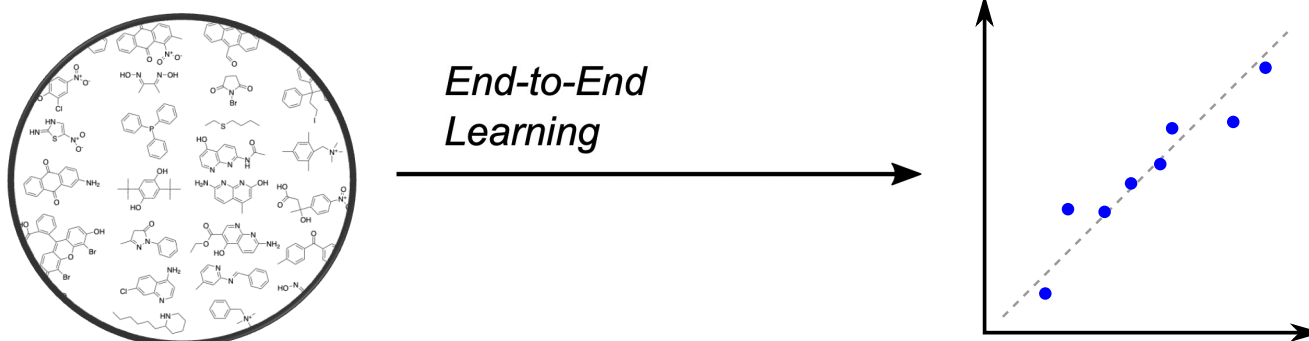
Traditional Cheminformatics Approach



Traditional cheminformatics / machine learning

- Determine a scheme to map atoms/bonds/molecules to a continuous vector space
- Fit a function that maps the vector space to desired property

Deep Learning Approach



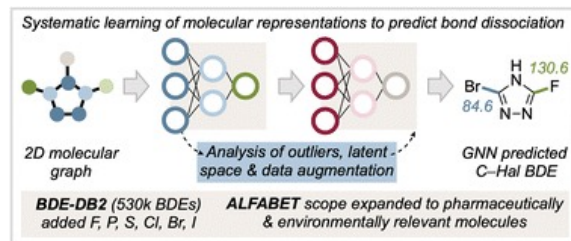
Traditional cheminformatics / machine learning

- Determine a scheme to map atoms/bonds/molecules to a continuous vector space
- Fit a function that maps the vector space to desired property

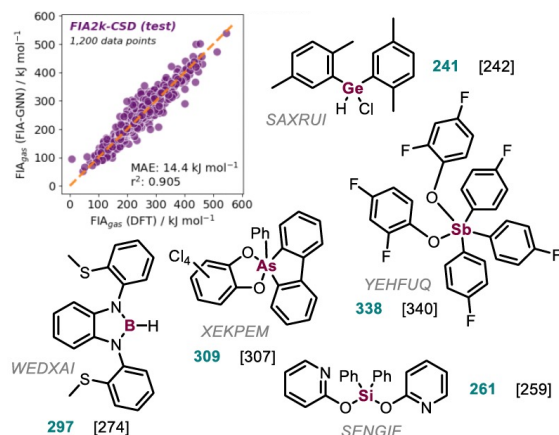
- Instead of manually determining a method to convert molecules into a finite-dimensional numerical representation, can we "learn" an optimal embedding?

Thermochemical and Spectroscopic Prediction Models

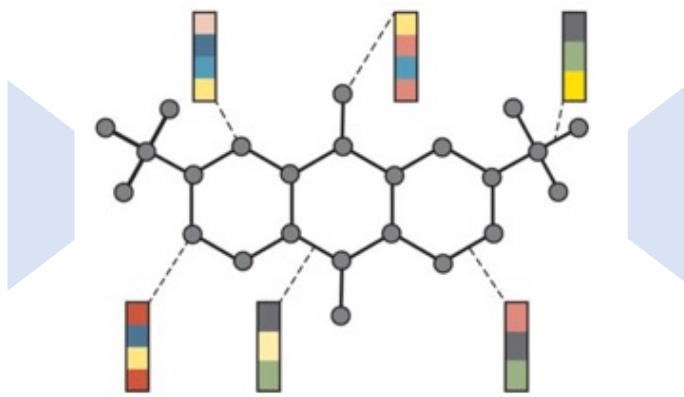
Reaction Thermochemistry



BDFE/BDE prediction: *Nat. Commun.* 2020, *Digit Discov* 2023

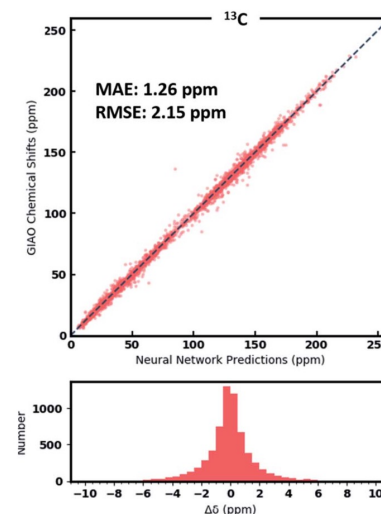


Lewis Acidity prediction: *ACIE* 2024
(with Lutz Greb, Heidelberg)



Graph Neural Network-based predictions

Spectroscopy



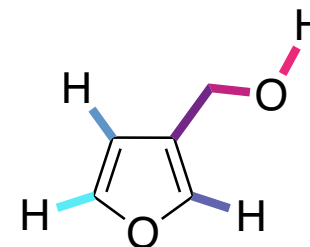
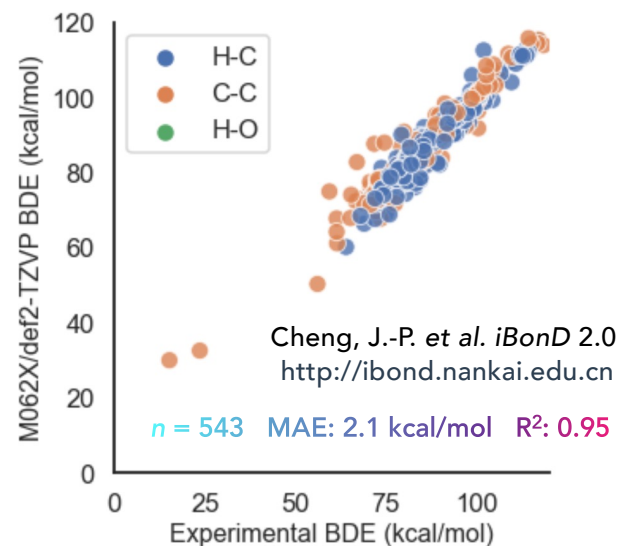
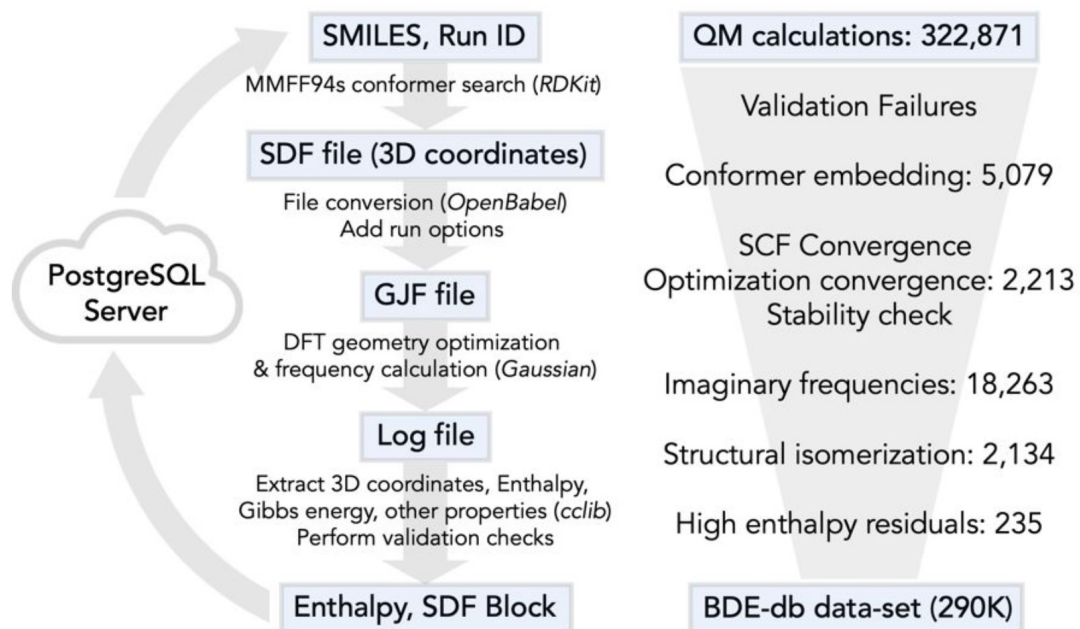
¹³C/¹H NMR : *Chem. Sci.* 2021

Solvation Free Energies

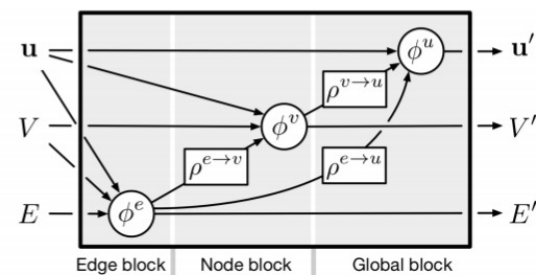
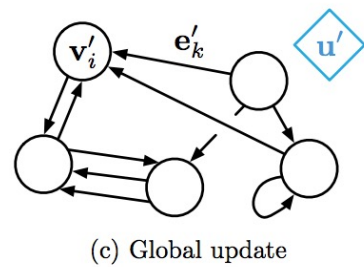
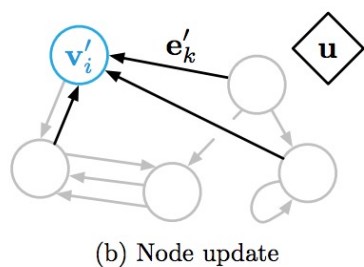
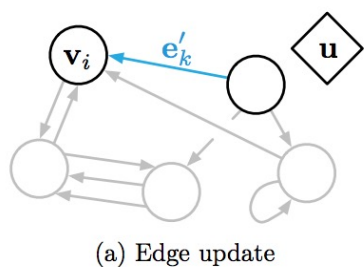
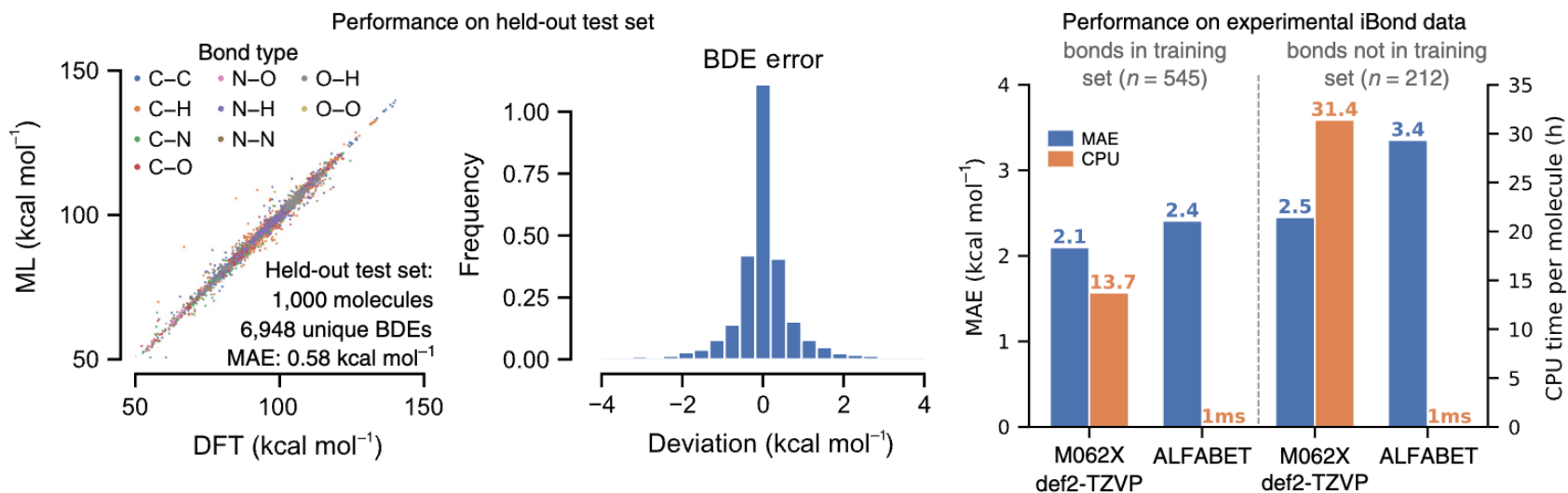


ΔG_{solv} : *Chem. Sci.* 2024
(with Kim, CSU)

Quantum Chemistry Workflow for 250K Species



An ML derived, Fast, Accurate Bond Dissociation Enthalpy Tool (ALFABET)



Battaglia PW, et al. *arXiv*. 2018;cs.LG

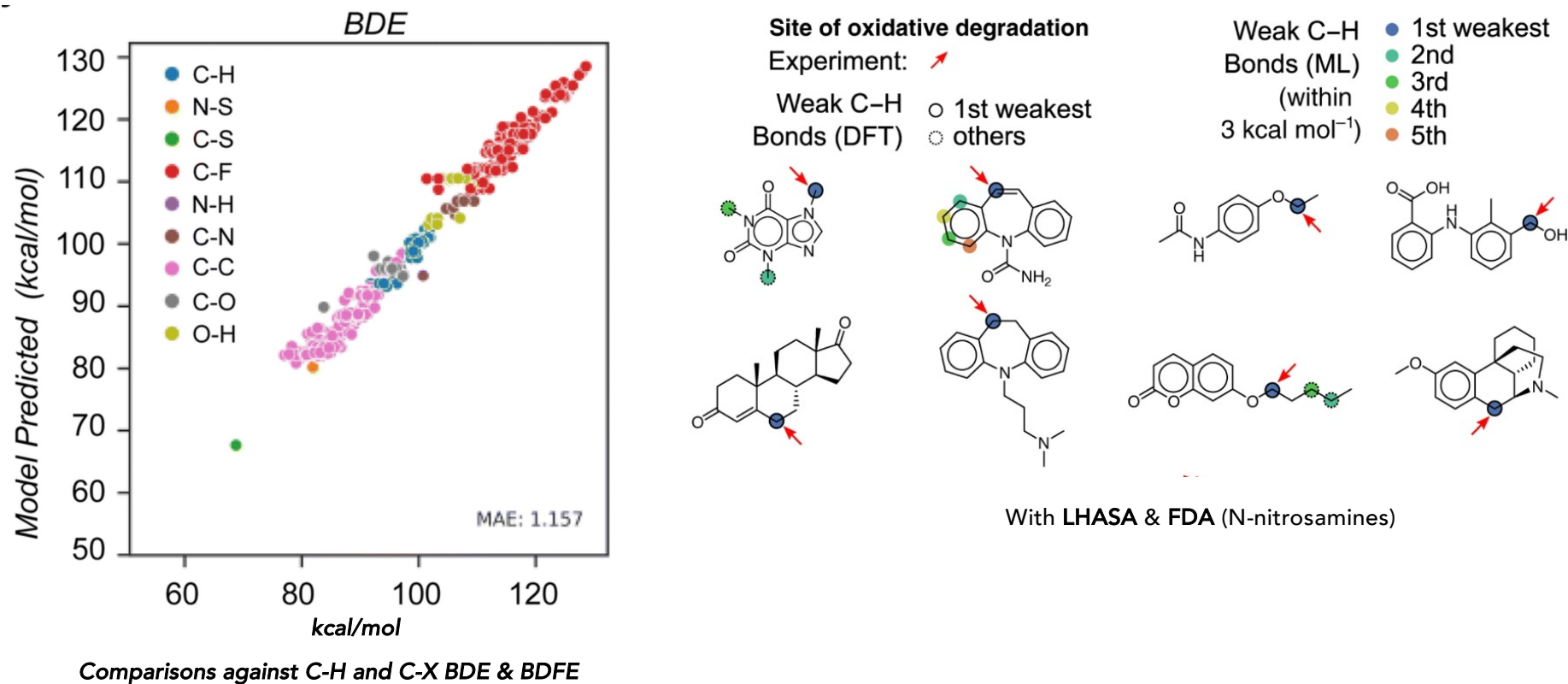
St. John, Guan, Kim, Kim & Paton *Nat. Commun.* 2020, 11, 2328

Code: <https://github.com/NREL/nfp>

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An ML derived, Fast, Accurate Bond Dissociation Enthalpy Tool (**ALFABET**)

Applications to Reactivity and Site-Selectivity (thermodynamics only!)



Applications of ALFABET

Webserver: https://huggingface.co/spaces/patonlab/alfabet_bde



A machine-Learning derived, Fast,
Accurate Bond dissociation Enthalpy Tool

Enter a SMILES string, e.g. 'CC1=CC(=CC(=C1)O)C'

DRAW

SUBMIT

Machine learning predictions of bond dissociation energies (BDEs)

This tool predicts BDEs for single, noncyclic bonds in neutral organic molecules consisting of C, H, O, and N atoms. Mean absolute errors are typically less than 1 kcal/mol for most compounds. To use, enter a SMILES string above (or use the drawing tool) and press submit. Reference DFT-calculated BDEs used as training can be displayed for any predicted bond using the neighbors link.

For more info, see our publications:

St. John, P.C., Guan, Y., Kim, Y., Kim, S., and Paton, R.S. Prediction of organic homolytic bond dissociation enthalpies at near chemical accuracy with sub-second computational cost. *Nat Commun* **11**, 2328 (2020).

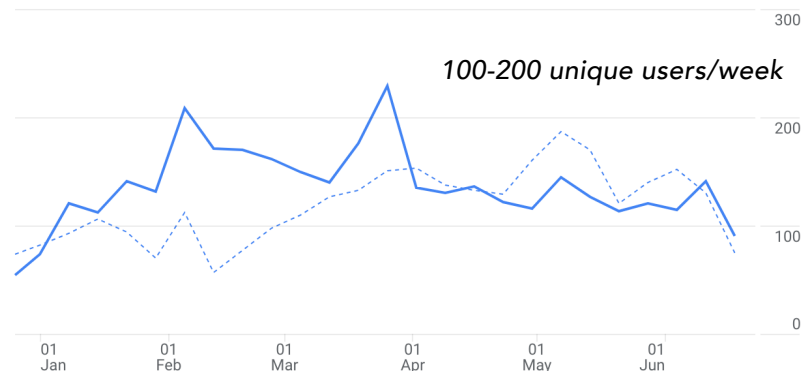
<https://doi.org/10.1038/s41467-020-16201-z>

St. John, P.C., Guan, Y., Kim, Y., Etz, B.D., Kim S., and Paton, R.S. Quantum chemical calculations for over 200,000 organic radical species and 40,000 associated closed-shell molecules. *Sci Data* **7**, 244 (2020). <https://doi.org/10.1038/s41597-020-00588-x>

pip install alfabet

```
>>> from alfabet import model
>>> model.predict(['CC', 'NCCO'])
```

	molecule	bond_index	bond_type	fragment1	fragment2	...	bde_pred
0	CC	0	C-C	[CH3]	[CH3]	...	90.278282
1	CC	1	C-H	[H]	[CH2]C	...	99.346184
2	NCCO	0	C-N	[CH2]CO	[NH2]	...	89.988495
3	NCCO	1	C-C	[CH2]O	[CH2]N	...	82.122429
4	NCCO	2	C-O	[CH2]CN	[OH]	...	98.250961
5	NCCO	3	H-N	[H]	[NH]CCO	...	99.134750
6	NCCO	5	C-H	[H]	N[CH]CO	...	92.216087
7	NCCO	7	C-H	[H]	NC[CH]O	...	92.562988
8	NCCO	9	H-O	[H]	NCC[O]	...	105.120598



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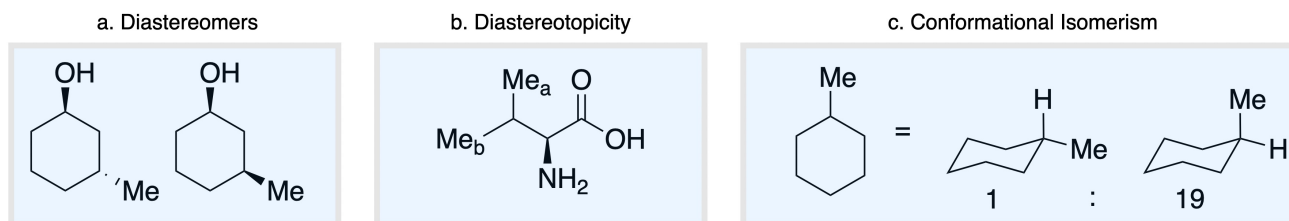
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St. John, Guan, Kim, Kim & Paton *Nat. Commun.* **2020**, *11*, 2328

Chemical Shift Calculation using DEep Learning (CASCADE)

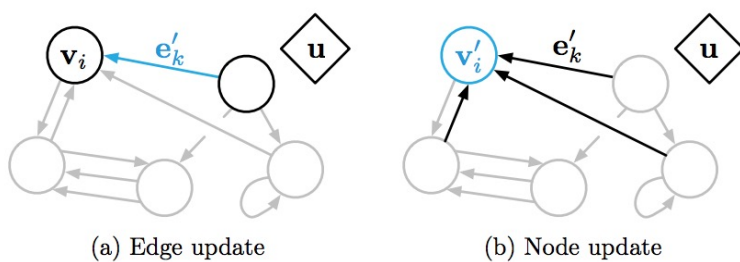
- NMR chemical shifts are much more sensitive to stereochemistry and conformation than BDE values



Dr. Yanfei Guan
(Takeda)

2D vs. 3D representation?

- Diastereotopic groups are not easily distinguished in a 2D molecular graph representation
- Conformational ensemble weights are not fully intrinsic to the molecule – e.g. solvent, temperature dependence
- Slow and fast exchange result in qualitatively different spectra



- Interatomic distances are rotationally-invariant
- They are expanded as a series of Gaussian radial basis functions
Schütt, K. T. et al. *J. Chem. Phys.* **2018**, 148, 241722
- Empirically we found a cutoff of 5Å worked best

ChemicAl Shift CA l culation using DEep Learning (CASCAD E)



Abhijeet
Bhaduria

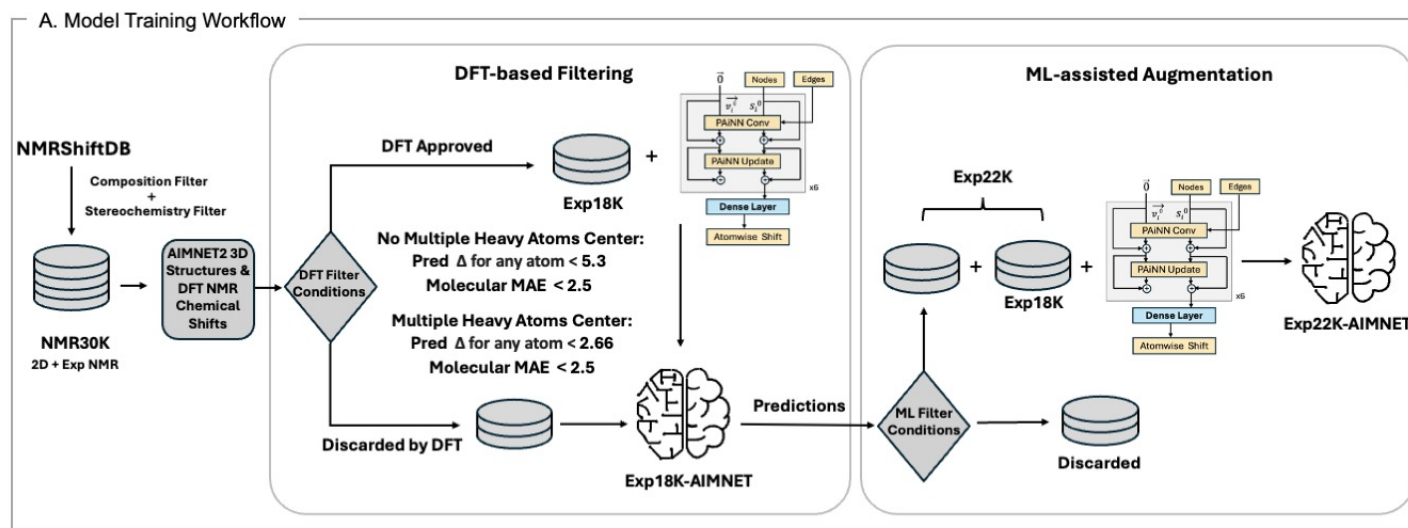


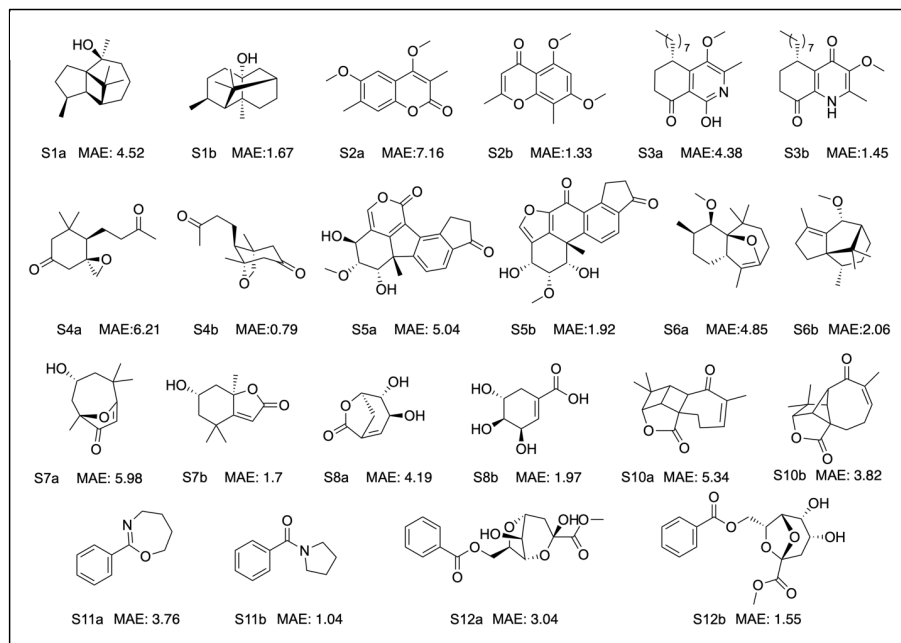
Table-2: CASCADE model performance improvement with dataset expansion

Model	Input Geometry	Dataset-Shifts	MAE (ppm)
ExpNN-DFT-2.0	DFT	Exp5K	1.08
Exp18K-AIMNET	AIMNET	Exp18K	0.76
Exp22K-AIMNET	AIMNET	Exp22K	0.73
Exp18K-FF	AIMNET	Exp18K	0.81
Exp22K-FF	AIMNET	Exp22K	0.78
FullSSPrUCe	MMFF	NMRShiftDB-Exp (33K)	1.22

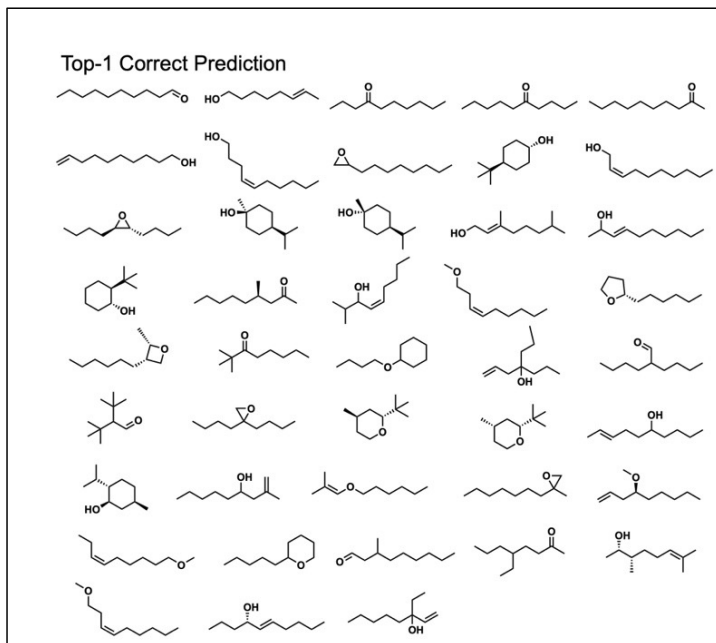
ChemicAI Shift Calculation using DEep Learning (CASCADE)

Webserver: <https://huggingface.co/spaces/patonlab/cascade>

real-world structure reassignment examples



$C_{10}H_{20}O$: top 1 43/50; top 3 50/50



Dr. Mihai
Popescu
(Vienna)

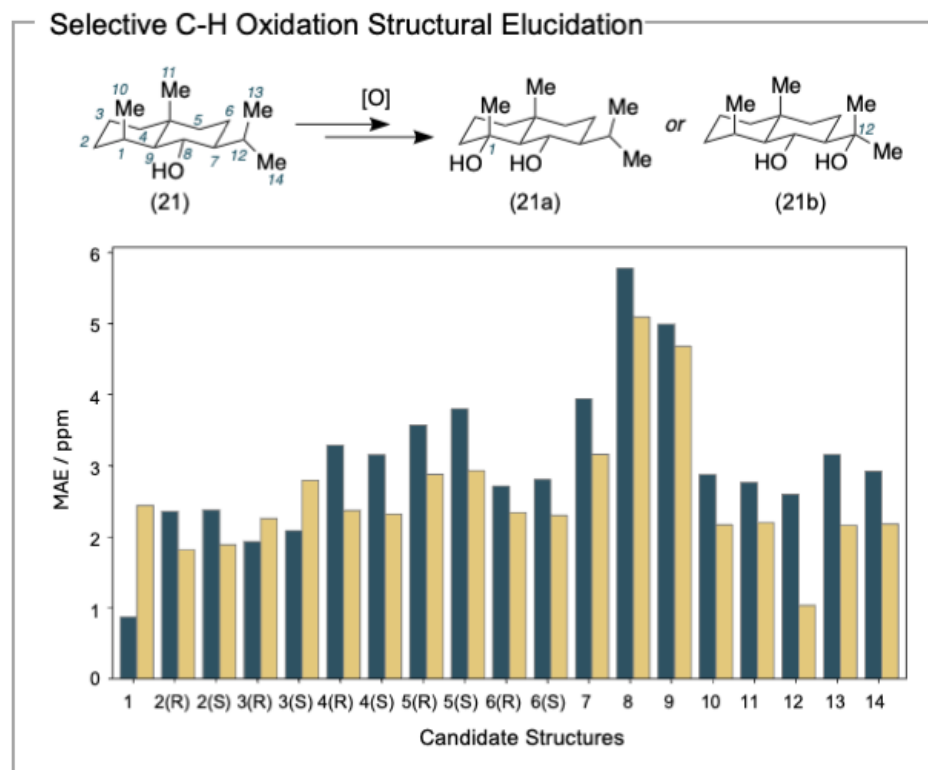
*based on systematic enumeration and ranking
of 32,677 possible structural and stereoisomers*

ChemicAI Shift CAIculation using DEep Learning (CASCADE)

Product determination for site- and stereo-selective C-H functionalizations

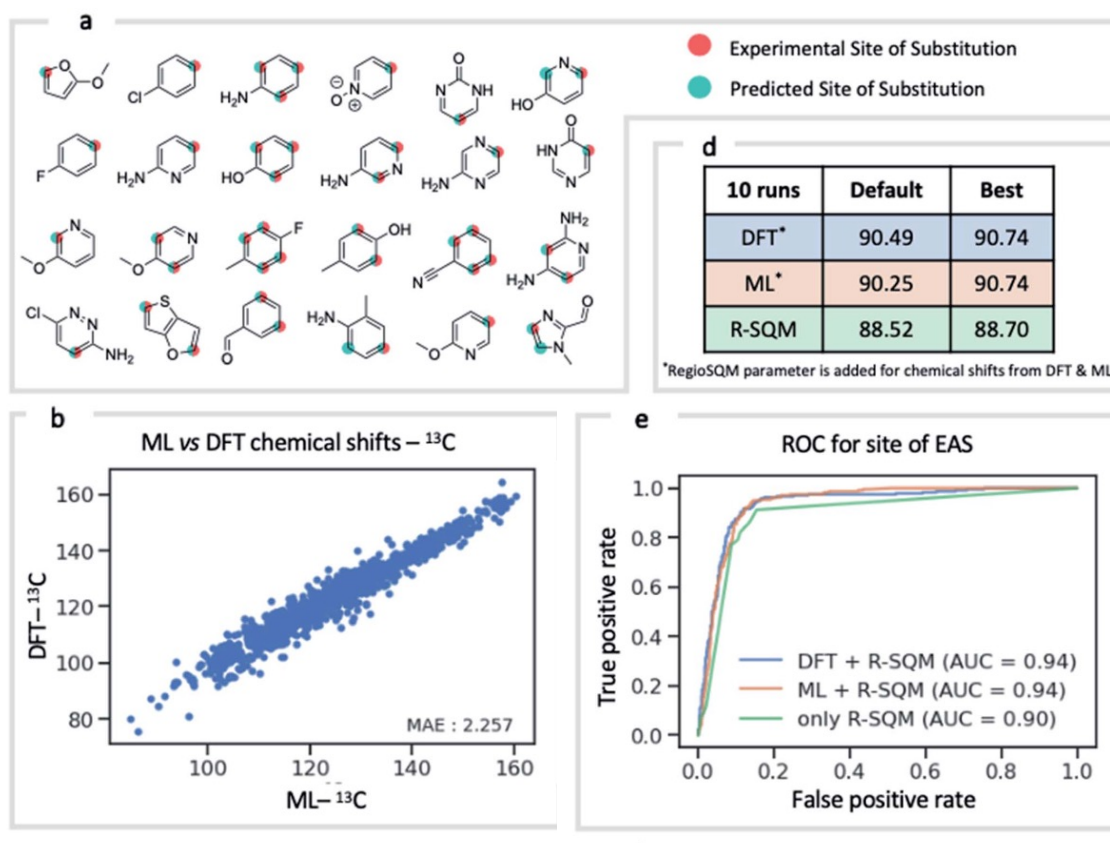


Abhijeet
Bhaduria



Reaction Data: Baran: Baran, Nature. 2009, 459, 824

Rapid Prediction of Regioselectivity (EAS)



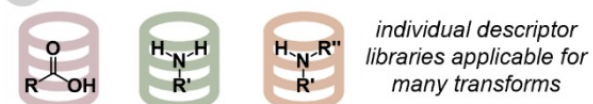
Dr. Liliana Gallegos
CMU – Gomes/Isayev

Site-selectivity data: Norrby *J. Org. Chem.* **2018**, *84*, 4695

Guan, Sowndarya, Gallegos, St. John & Paton *Chem. Sci.* **2021**, *12*, 12012-12026

Rapid Prediction of Reactivity: Amide Coupling

A. Amide Coupling Case Study



previous work:



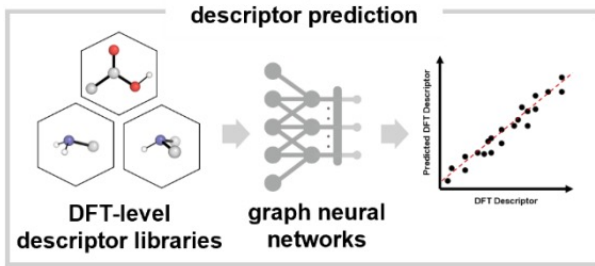
44 reaction rates correlated to DFT-level descriptors

B. Descriptor Prediction Workflow

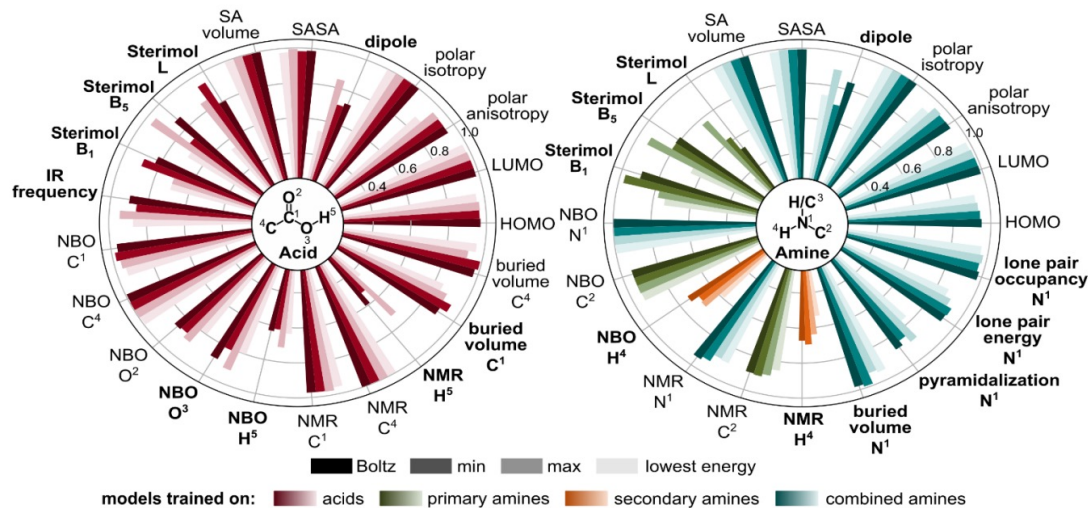
user input:



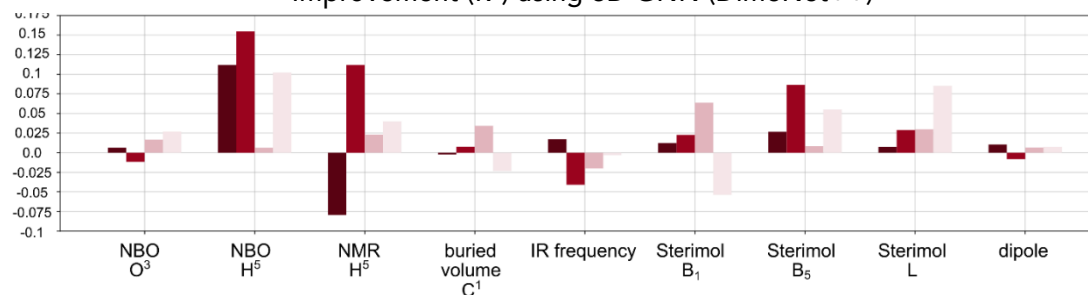
descriptor prediction



Accuracy of 2D Graph Neural Networks



Improvement (R^2) using 3D GNN (DimeNet++)



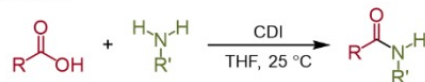
With Coley (MIT), Sigman (Utah): *Digit Discov* **2025**, 4, 222-233

Rapid Prediction of Reactivity: Amide Coupling

A. Amide Coupling Case Study



previous work:



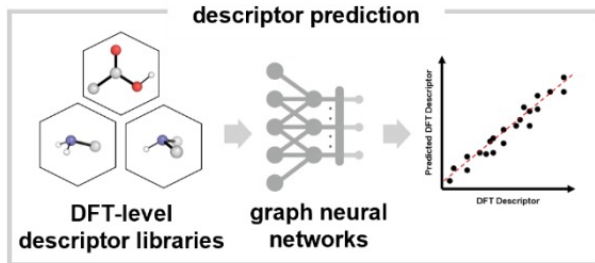
44 reaction rates correlated to DFT-level descriptors

B. Descriptor Prediction Workflow

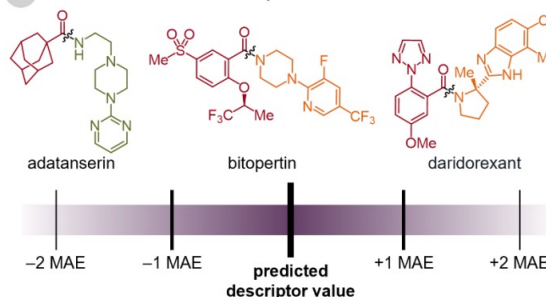
user input:



descriptor prediction

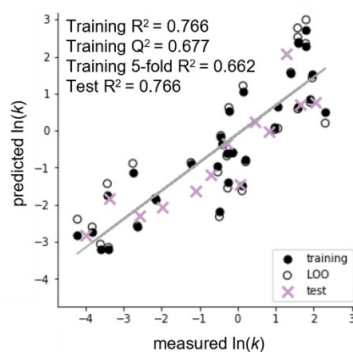


A. External Validation of Descriptor Prediction Models



B. Rebuilt Amide Coupling Rate Model

$$\ln(k) = -0.37 - 0.76 \text{ acid}\%V_{\text{bur } 3.0\text{\AA}}(\text{C1}) - 0.95 \text{ acidNBO}_{(\text{C1})} + 1.16 \text{ acid}V_{(\text{C1}=\text{O}2)} - 0.71 \text{ amineSterimol } B_1(\text{N1} \dots \text{C2})$$



retrained models for virtual screening

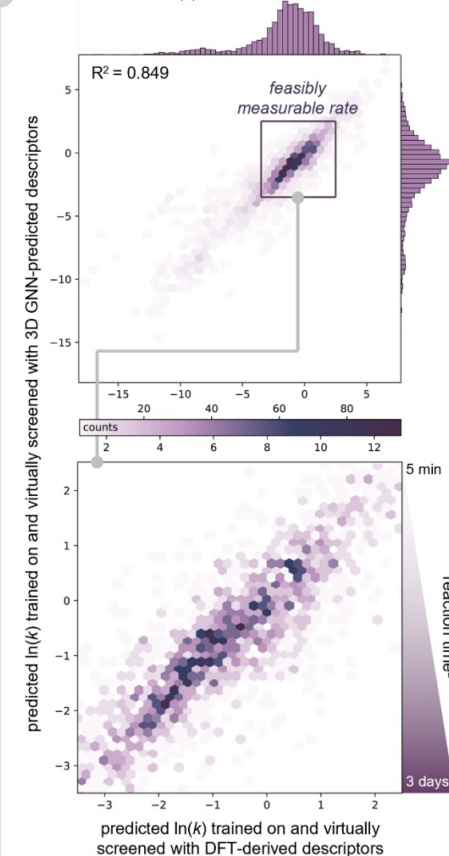
3D GNN-predicted

$$\ln(k) = -0.44 - 0.89 \text{ acid}\%V_{\text{bur } 3.0\text{\AA}}(\text{C1}) - 0.91 \text{ acidNBO}_{(\text{C1})} + 1.01 \text{ acid}V_{(\text{C1}=\text{O}2)} - 0.81 \text{ amineSterimol } B_1(\text{N1} \dots \text{C2})$$

DFT-derived

$$\ln(k) = -0.44 - 0.84 \text{ acid}\%V_{\text{bur } 3.0\text{\AA}}(\text{C1}) - 0.93 \text{ acidNBO}_{(\text{C1})} + 0.99 \text{ acid}V_{(\text{C1}=\text{O}2)} - 0.84 \text{ amineSterimol } B_1(\text{N1} \dots \text{C2})$$

C. Error in Predicted $\ln(k)$



With Coley (MIT), Sigman (Utah): *Digit Discov* **2025**, 4, 222-233

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